MIL-I-38391(USAF)
AMENDMENT 3
18 Mar 86
SUPERSEDING
AMENDMENT 2
15 MARCH 1966

MILITARY SPECIFICATION

INDICATOR, ELECTRICAL TACHOMETER,
ERU-12/A 0-350 RPM (ROTOR), 0-700
RPM (ENGINE)

This amendment forms a part of MIL-I-38391(USAF), dated 5 May 1965, and is approved for use within the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

PAGE 1

* 1.1 Add title "Scope".

* Delete 2.1 and add:
"2.1 Government documents".

* Add:
"2.1.1 Specifications standards, and handbooks. The following specifications, standard, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATION, Federal delete specifications "PPP-B-566, PPP-B-665 and PPP-B-676" and add: "PPP-B-601, Boxes, Wood, Cleated Plywood."

* Under SPECIFICATION, Military delete "MIL-P-7936, MIL-M-26512, MIL-R-26667 and MIL-D-70327" and add "DOD-D-100 Drawing, Engineering and Associated List".

* Under STANDARDS, Military delete "MS33586" and add "MIL-STD-470 Maintainability Program Requirements (For Systems and Equipments)", "MIL-STD-471 Maintainability Demonstration", "MIL-STD-781 Reliability Tests Exponential distribution", "MIL-STD-794 Part and Equipment, Procedures For Packaging Of" and "MIL-STD-889 Dissimilar Metals".

* Add boxed paragraph at bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Propsal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3.3 Line 2 delete "MS33586" and add "MIL-STD-889".

* Add:
"3.3.4.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

* 3.4.3 Line 2 delete "MIL-M-26512" and add "MIL-STD-470".

PAGE 4

Table I: In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE", delete "20" (three places) and substitute "30" and delete "10" (one place) and substitute "20".

PAGE 5

Table II: In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE, delete "20" (three places) and substitute "30" and delete "10" (one place) and substitute "20".

PAGE 6

Table III: In column entitles "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE opposite generator drive shaft speed 4100, delete "1" and substitute "2".

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Table IV: In column entitled "SCALE ERROR TOLERANCE ± ROOM TEMPERATURE," opposite generator drive shaft speed 4117, delete "1" and substitute "2".

PAGE 8

* 3.6 Line 6 delete "MIL-D-70327" and add "DOD-D-100".

3.7 Fourth line: Delete "5.00 inches" and add "6.00 inches".

PAGE 15

Add:
"4.3.9 Order of tests. Unless otherwise specified, tests shall be conducted in the order shown under 4.6."

PAGE 16

Paragraph 4.5.2.2: Delete the test listing and substitute:

"a. Sampling plan A tests
b. Position error
c. Magnetic effect
d. Indicator current
e. Overspeed and reverse operation
f. Thermal shock
g. Vibration failure

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h. Endurance
i. Salt spray

Note: See 4.3.8 for order of testing."

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Paragraph 4.6.2: Add as second and third sentences: "The indicators shall be operated until their internal temperature has stabilized. For this test, the indicators shall be warmed up a minimum of 20 minutes. THIS WARMUP REQUIREMENT WILL NOT APPLY WHEN THE SCALE ERROR TEST IS APPLICABLE TO 4.6.7 AND 4.6..8".

Paragraph 4.6.5: Delete the second sentence and substitute: "Within 10 seconds after removal from the oven, the indicator cover glass shall be immersed to a depth of 1/4 inch in an ice-water bath maintained at 0° to 5°C." In the third sentence, delete "3 minutes" and substitute "5 minutes".

PAGE 18

Paragraph 4.6.6: Add as last sentence: "Evidence of fogging or damage shall be caused for rejection".

PAGE 20

Paragraph 4.6.20 : Delete "MIL-R-26667" and substitute "MIL-STD-781".

PAGE 21

Paragraph 4.6.20.2: Delete and substitute:
"4.6.20.2 Test level. The test level shall be in accordance with MIL-STD-781, test level F except that the indicators will be operated continously at a generator drive shaft speed of 3000 - 4000 rpm and the low temperature portion of the temperature cycling shall be conducted at -35°C. Input voltages need not be monitored."

Paragraph 4.6.20.3.2: Delete "MIL-R-26667" and substitute "MIL-STD-781".

* 4.6.21 Line 3 delete "MIL-M-26512" and add "MIL-STD-471".

* Add a new paragraph to Section 4 to read as follows:
"4.7 Inspection and test. Inspection and test of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

* Delete Section 5 in its entirety and substitute the following:
"5 PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Indicators shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4. Unit packaging. Unless otherwise specified by the contracting activity, each indicator, electrical tachometer, ERU-12/A 0-350 RPM (Rotor), 0-7000 RPM (Engine), shall be packaged in quantity unit packs of one each in accordance with Method 1A of MIL-P-116. Each unit shall be overboxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock tranmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Indicators shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, style A or B, Class overseas, unless otherwise specified of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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